Model Development Phase Template

Date	27 August 2026
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Project Title	Fetal Health Classification System
Maximum Marks	10 Marks

Initial Model Training Code, Model Validation and Evaluation Report

Initial Model Training Code (5 marks):

The initial model training is implemented in train_model.py, loading the pre processed CTG dataset (fetal_health.csv) and evaluating three models: Logistic Regression, Random Forest, and XGBoost. The code includes data preprocessing, model training, and evaluation.

```
from matplotlib.pylab import logistic
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import classification_report
import joblib
 # Load dataset
df = pd.read_csv(data/fetal_health.csv)
X = df.drop(fetal_health, axis=1)
y = df[fetal_health]
# Standardize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
joblib.dump(scaler, scaler.pkl)
 # Adjust target for XGBoost
y_adjusted = y- 1
 # Split data
X_train, X_test, y_train, y_test = train_test_split(X_scaled,
y_adjusted, test_size=0.2, random_state=42, stratify=y_adjusted)
 # Initialize models
models = {
logistic_Regression: LogisticRegression(multi_class=
multinomial, max_iter=1000, class_weight=balanced),
Random_Forest: RandomForestClassifier(random_state=42,
class_weight=balanced),
XGBoost: XGBClassifier(use_label_encoder=False, eval_metric=
mlogloss, random_state=42)
 }
 # Train and evaluate
for name, model in models.items():
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(f"{name}_Performance:")
print(classification_report(y_test, y_pred, target_names=[
Normal, Suspect, Pathological]))
 # Save XGBoost model
joblib.dump(models[XGBoost],fetalai_model.pkl)
```

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Model Validation and Evaluation Report (5 marks):

Model	Summary	Training and Validation Performance Metrics
Logistic Regression	Linear model with multinomial loss and balanced class weights. Parameters: 21 features × 3 classes.	**Training Accuracy: ~82% Test Accuracy: ~78% F1-Score (macro): ~0.75 Limited performance on minority classes.**
Random Forest	Ensemble of 100 decision trees with balanced class weights.	**Training Accuracy: ~95% Test Accuracy: ~85% F1-Score (macro): ~0.80 Better handling of class imbalance.**
XGBoost	Gradient boosting with tree-based learning. Parameters: Default settings.	**Training Accuracy: ~98% Test Accuracy: ~90% F1-Score (macro): ~0.85 Best performance across all classes.**

Conclusion :

The XGBoost model achieved the highest performance with a test accuracy of ~90% and a macro F1-score of ~0.85, outperforming Logistic Regression and Random Forest. Its robustness to class imbalance and ability to capture com plex feature relationships make it the optimal choice for FetalAI's fetal health classification task.